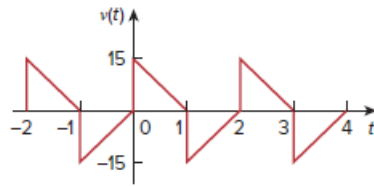


## Problem

Obtain the trigonometric Fourier series for the voltage waveform shown in Fig. 17.66.

**Figure 17.66:**



## Step-by-step solution

## Step 1 of 4

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} -15(t-1), & 0 < t < 1 \\ 15(t-2), & 1 < t < 2 \end{cases}$$

## Step 2 of 4

The time period is,  $T = 2$ .

Determine the value of  $\omega_0$ .

$$\begin{aligned}\omega_0 &= \frac{2\pi}{2} \\ &= \pi\end{aligned}$$

Refer to Equation 17.33 in the textbook.

Since the function is half-wave (odd) symmetric,

$$a_0 = 0$$

Determine the value of  $a_n$ .

$$\begin{aligned}a_n &= \frac{4}{T} \int_0^{T/2} f(t) \cos(n\omega_0 t) dt \\ &= \frac{4}{2} \int_0^1 f(t) \cos(n\omega_0 t) dt \\ &= 2 \int_0^1 -15(t-1) \cos(n\pi t) dt \\ &= \frac{1}{n\pi} \left[ -30(t-1) \sin(n\pi t) \right]_{t=0}^{t=1} - \frac{30}{n^2 \pi^2} \left[ \cos(n\pi t) \right]_{t=0}^{t=1} \\ &= 0 + \frac{30}{n^2 \pi^2} (1 - \cos(n\pi)) \\ &= \frac{30}{n^2 \pi^2} (1 - \cos(n\pi)) \\ &= \frac{60}{n^2 \pi^2}, n \text{ is odd}\end{aligned}$$

### Step 3 of 4

Determine the value of  $b_n$ .

$$\begin{aligned}b_n &= \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt \\ &= \frac{4}{2} \int_0^1 f(t) \sin(n\omega_0 t) dt \\ &= 2 \int_0^1 -15(t-1) \sin(n\pi t) dt \\ &= \frac{30}{n\pi} \left[ (t-1) \cos(n\pi t) \right]_{t=0}^{t=1} - \frac{30}{n^2 \pi^2} \left[ \sin(n\pi t) \right]_{t=0}^{t=1} \\ &= \frac{30}{n\pi} [0+1] - \frac{30}{n^2 \pi^2} [0-0] \\ &= \frac{30}{n\pi}\end{aligned}$$

### Step 4 of 4

The Fourier series is,

$$\begin{aligned}f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\&= 0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\&= \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \frac{60}{n^2 \pi^2} \cos(n\pi t) + \frac{30}{n\pi} \sin(n\pi t) \right\}\end{aligned}$$

Thus, the Fourier series of the periodic function is,

$$f(t) = \sum_{\substack{n=1 \\ n=\text{odd}}}^{\infty} \left\{ \frac{60}{n^2 \pi^2} \cos(n\pi t) + \frac{30}{n\pi} \sin(n\pi t) \right\}.$$